

MagMat-Discovery

Physics-Informed Machine Learning and Screening for Magnetic Materials

System Architecture, Methodology & Model Validation

Technical Report & Model Guide

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A practical guide to predicting magnetic phases, critical temperatures, and hysteresis properties using physics-informed machine learning, calibrated uncertainty ranges, and interactive materials screening.

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Executive Summary

Discovering high-performance permanent magnets with reduced rare-earth content is important for electric vehicles, wind turbines, and industrial machinery. However, magnetic materials are challenging to model because different magnetic properties are governed by different physical mechanisms across atomic, crystallographic, and microstructural length scales. Some magnetic properties, such as the Curie temperature (T_C) and Néel temperature (T_N), correlate well with chemical composition. Other properties, such as coercivity (H_c) and maximum energy product ($(BH)_{\max}$), depend strongly on crystal symmetry, local anisotropy, and microstructure, making them difficult to predict from composition alone.

MagMat-Discovery is an open machine learning and screening platform developed to predict both intrinsic and extrinsic magnetic properties. The platform integrates:

- **Curated Experimental Data:** A standardized dataset of 51,394 experimental entries from the Non-Equilibrium Magnetic Materials Database (NEMAD), linked to crystallographic symmetries from the Materials Project.
- **Physics-Guided Descriptors:** Featurization encoding valence electron concentration, superexchange transfer factors, spin-orbit proxies, and single-ion anisotropy terms.
- **Learned Representations:** A stoichiometric graph transformer (**MagFormer**) and equivariant structural embeddings from MACE.
- **Physical Cascades:** Enforcing thermodynamic ceilings and micromagnetic relationships so that downstream hysteresis predictions remain physically consistent.
- **Calibrated Uncertainty Ranges:** Conformal prediction intervals that provide an expected 90% coverage for every continuous prediction.

Across 8 magnetic target properties evaluated under leak-free grouped cross-validation, the platform achieves strong performance on magnetic phase classification (91.9% accuracy) and transition temperatures ($R^2 = 0.82$ for T_C , $R^2 = 0.74$ for T_N). For extrinsic hysteresis properties (H_c , K_1 , $(BH)_{\max}$), the models achieve moderate predictive performance ($R^2 = 0.46$ – 0.62), providing an effective ranking tool to filter promising candidates before detailed synthesis.

Keywords: Magnetic Materials, Physics-Informed Machine Learning, Graph Transformers, Conformal Calibration, Coercivity, Permanent Magnets, Materials Screening.

Glossary of Key Terms

To assist readers from different materials science and computational backgrounds, Table 1 summarizes the primary magnetic properties and machine learning terminology used throughout this report.

Table 1: Glossary of magnetic properties, physical units, and computational terms.

Term	Typical Unit	Physical Meaning / Definition
T_C (Curie Temp.)	K	Temperature above which a ferromagnetic material loses its spontaneous magnetization.
T_N (Néel Temp.)	K	Temperature above which an antiferromagnetic material becomes paramagnetic.
M_s (Saturation Mag.)	emu/g or T	Maximum magnetization achieved in an applied magnetic field. Reported in mass units (emu/g) or magnetic polarization $\mu_0 M_s$ (T).
H_c (Coercivity)	A/m or kA/m	Applied magnetic field required to reduce magnetization back to zero after saturation.
K_1 (Anisotropy Constant)	J/m ³	Energy density required to rotate magnetization away from the preferred crystal axis.
$(BH)_{\max}$ (Energy Product)	kJ/m ³	Primary figure of merit for permanent magnets, representing stored magnetic energy density.
θ_p (Paramagnetic Curie Temp.)	K	High-temperature intercept from Curie-Weiss susceptibility fitting; its sign reflects dominant exchange coupling.
FM / AFM / NM	Discrete	Ground-state magnetic ordering: Ferromagnetic (parallel), Antiferromagnetic (antiparallel), or Non-magnetic.
Out-of-Fold (OOF)	–	Predictions made on data held out during cross-validation, ensuring unbiased inputs to downstream models.
Conformal Range	Physical	A calibrated uncertainty interval constructed from validation residuals, expected to contain the true value approximately 90% of the time.
dex	log ₁₀	Decimal exponent (order of magnitude). An error of 0.65 dex means a factor of $10^{0.65} \approx 4.5$.

1 Project Background and Objectives

1.1 Motivation and Practical Needs

Permanent magnets are critical components in modern clean energy and transportation systems, including electric vehicle traction motors, wind turbine generators, and precision actuators [6]. The strongest commercially available magnets rely heavily on critical rare-earth elements, primarily Neodymium (Nd), Dysprosium (Dy), and Samarium (Sm). These elements are vulnerable to supply disruptions, price volatility, and environmentally damaging mining practices.

Developing alternatives with lower rare-earth content—or eliminating rare earths entirely—requires materials that meet three criteria:

1. Elevated Curie temperature ($T_C > 400\text{ K}$) to prevent thermal demagnetization during operation.
2. Substantial saturation magnetization ($\mu_0 M_s > 1.0\text{ T}$) to ensure strong flux output.
3. High coercivity ($H_c > 800\text{ kA/m}$) to resist opposing demagnetizing fields.

1.2 Why Machine Learning for Magnetism is Hard

Applying machine learning to magnetic materials presents specific challenges that distinguish it from standard property prediction tasks:

- **Some properties are harder to predict than others:** Chemical composition alone correlates well with Curie and Néel temperatures because magnetic exchange is largely set by local chemical bonds. In contrast, coercivity (H_c) and anisotropy (K_1) depend strongly on crystal symmetry, grain boundaries, and defect pinning. Composition-only models typically struggle on these extrinsic targets.
- **Standard data splits can produce misleading metrics:** Experimental databases frequently contain many measurements of the same parent material with minor dopant substitutions. Standard random splits place identical chemical formulas in both training and test sets. This creates artificial memorization and inflates validation scores, while failing on genuinely new chemistries.
- **Independent models can violate physical laws:** When target properties are predicted independently, statistical models can produce contradictory results. For example, a model might predict an energy product $(BH)_{\max}$ that exceeds the theoretical ceiling ($\frac{1}{4}\mu_0 M_s^2$), or predict non-zero coercivity for a non-magnetic compound.

1.3 Objectives of this Work

MagMat-Discovery was built to address these challenges through four clear objectives:

1. **Unified Multi-Target Modeling:** Simultaneously predict 8 magnetic properties covering magnetic phase, transition temperatures, and hysteresis behavior.
2. **Leak-Free Grouped Evaluation:** Evaluate all models using 5-fold cross-validation grouped by reduced chemical formula, ensuring identical compositions do not appear in both train and test sets.
3. **Physics-Enforced Cascades:** Connect intrinsic predictions (K_1, M_s) to extrinsic predictions ($H_c, (BH)_{\max}$) using micromagnetic relationships and thermodynamic bounds.
4. **Calibrated Ranges and Interactive Screening:** Provide 90% conformal uncertainty ranges for every prediction and package the pipeline into an accessible web platform.

2 Data Curation and Split Design

2.1 Experimental and Crystallographic Data Sources

MagMat-Discovery brings together experimental measurements and crystallographic metadata:

- **NEMAD Database:** 51,394 curated experimental records from the Non-Equilibrium Magnetic Materials Database [1]. These provide measured values for phase, T_C , T_N , M_s , K_1 , H_c , $(BH)_{\max}$, and θ_p .
- **Materials Project:** High-throughput crystal structures and symmetry labels [2], providing space group numbers (1–230), crystal systems, and densities.
- **Bilbao MAGNDATA:** Magnetic space group ground truths for magnetic symmetry validation.

2.2 Data Cleaning and Feature Sanitation

Prior to model training, the raw datasets undergo systematic cleaning:

1. **Exclusion of Unmeasurable Features:** DFT-computed total energy, formation energy, and calculated magnetic moments are intentionally excluded from input features. Because these quantities are unavailable for uncharacterized materials, including them would leak the ground-state phase.
2. **Formula Canonicalization:** Chemical formulas are parsed and normalized to coprime integer stoichiometries (`reduced_formula`) using `pymatgen`.
3. **Logarithmic Scaling for Wide-Span Targets:** Properties that span multiple orders of magnitude (H_c , K_1 , $(BH)_{\max}$) are transformed using base-10 logarithmic or symmetric-logarithmic scaling. This stabilizes variance during model training.

2.3 Grouped Data Partitioning and Leakage Controls

To evaluate model generalization honestly, cross-validation is performed using 5-fold `GroupKFold` grouped by `reduced_formula`.

This grouping guarantees that identical chemical formulas and pure polymorphs (such as different crystal phases of Fe_3Pt or repeated entries of $\text{Nd}_2\text{Fe}_{14}\text{B}$) never appear in both training and test folds simultaneously.

It is worth noting that chemically substituted variants with distinct reduced formulas (for example, $\text{Nd}_2\text{Fe}_{13}\text{CoB}$ versus $\text{Nd}_2\text{Fe}_{14}\text{B}$) are assigned to separate groups. This tests whether the models can generalize across neighboring chemical substitutions without memorizing the exact stoichiometry.

3 Methodology and System Architecture

3.1 Overview of the Pipeline

Figure 1 shows the complete pipeline: data ingestion, multi-tier physical featurization, graph transformer embeddings, ensemble modeling, and physical constraint enforcement.

3.2 Multi-Tier Physics Descriptors

Features are structured into four physical tiers. The underlying mathematical formulations are provided in Appendix A:

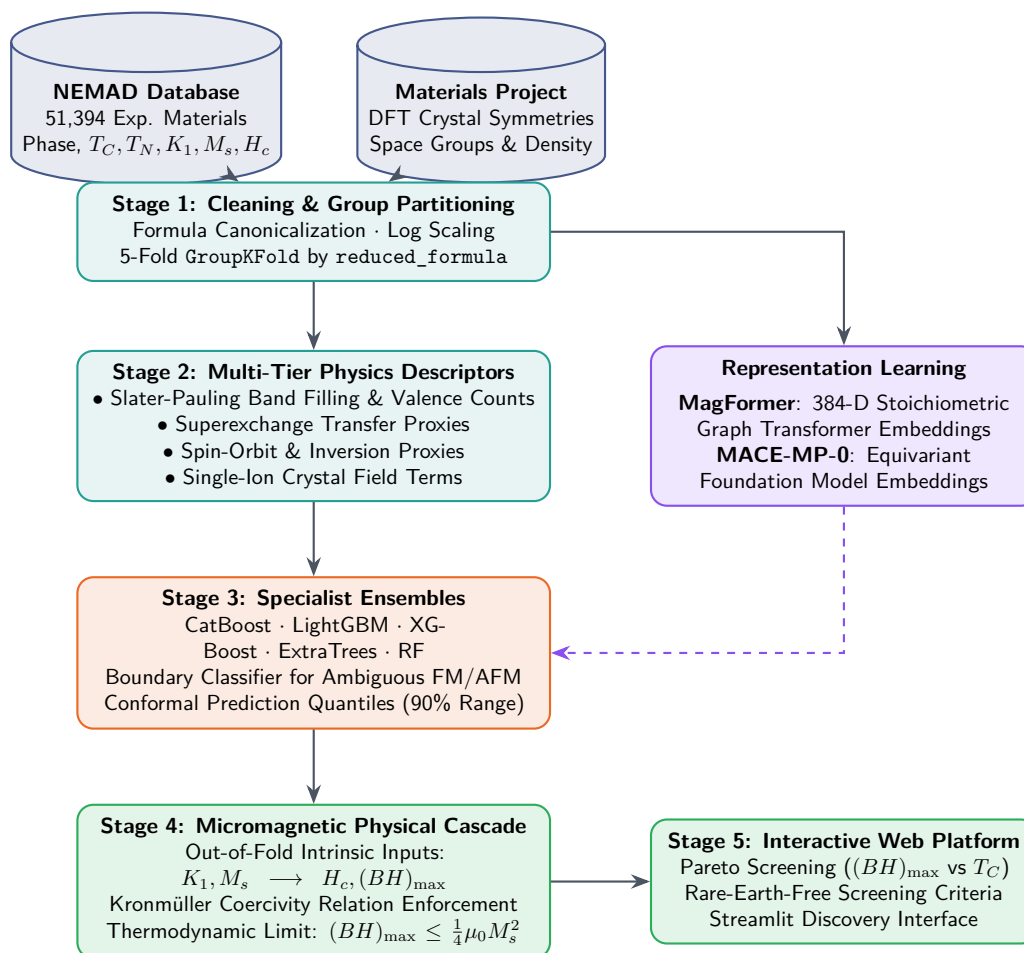


Figure 1: End-to-end pipeline flowchart of MagMat-Discovery. Data undergoes leak-free preprocessing, feeding both physics-guided descriptors and learned representations from MagFormer and MACE. Out-of-fold intrinsic predictions drive downstream micromagnetic cascades, conformal uncertainty calibration, and candidate screening.

- **Tier 1 (Stoichiometry and Elements):** Elemental mass fractions, atomic numbers, covalent radii, and Pauling electronegativities.
- **Tier 2 (Electronic Structure and Exchange Physics):** Valence electron concentration (VEC) relative to the Slater-Pauling maximum ($\text{VEC} \approx 8.3$); superexchange factors that model whether non-magnetic anions promote ferromagnetic or antiferromagnetic alignment; and spin-orbit proxies that account for heavy-element interactions.
- **Tier 3 (Crystal Field and Anisotropy):** Crystal system indicators and Stevens operator crystal-field terms that signal whether an alloy favors an easy-axis ($K_1 > 0$) or easy-plane orientation.
- **Tier 4 (Structural Foundation Embeddings):** Pretrained representations from the MACE-MP-0 equivariant model [3] for compounds with known crystal coordinates.

3.3 Deep Representation Learning with MagFormer

When screening hypothetical compositions, exact atomic coordinates are often unavailable. To extract learned chemical representations from formula strings alone, we developed **MagFormer**, a stoichiometric graph transformer.

MagFormer represents each chemical formula as a fully connected element graph. Constituents are initialized with elemental properties, and multi-head attention is modulated by pairwise property differences. Stoichiometry-weighted pooling produces a 384-dimensional representation vector for each compound. These vectors are passed as auxiliary features to the gradient-boosted ensembles. Detailed network specifications are provided in Appendix B.

3.4 Ensemble Regressors and Micromagnetic Cascades

Each target property is modeled using an ensemble of five diverse regressors: CatBoost, LightGBM, XGBoost, ExtraTrees, and Random Forest. Predictions are combined using non-negative least squares meta-blending. For ground-state phase identification, a dedicated boundary classifier is trained specifically on ambiguous compounds where predicted FM and AFM probabilities are close ($|P(\text{FM}) - P(\text{AFM})| < 0.20$).

To prevent unphysical predictions in extrinsic hysteresis targets, the pipeline applies a physical cascade:

1. Intrinsic properties (K_1 and M_s) are predicted first.
2. Their out-of-fold predictions are passed as inputs to the coercivity (H_c) and energy product ($(BH)_{\max}$) models. This informs the model of the Kronmüller coercivity relationship [4]:

$$H_c \approx \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (1)$$

where α_{micro} reflects microstructural reduction and N_{eff} is an effective demagnetizing factor.

3. Predicted energy products are bounded by the thermodynamic ceiling: $(BH)_{\max} \leq \frac{1}{4} \mu_0 M_s^2$.

3.5 Calibrated Uncertainty Estimation

Every continuous prediction is accompanied by a range. We apply split conformal prediction [5] using out-of-fold validation residuals. The 90th percentile of the absolute residuals defines a calibration margin:

$$q_{\text{conf}} = \text{Quantile}_{0.90}(\{|y_i - \hat{y}_i|\}_{i=1}^n) \quad (2)$$

For any new material, the prediction range is given by $[\hat{y} - q_{\text{conf}}, \hat{y} + q_{\text{conf}}]$. Under standard exchangeability, approximately 90% of true values fall within this range.

4 Model Validation and Performance Evaluation

4.1 How to Read the Evaluation Metrics

Before examining the benchmark results, it is helpful to clarify what each metric means:

- **Coefficient of Determination (R^2):** Measures the proportion of variance explained by the model. An R^2 of 1.0 represents perfect prediction, while an $R^2 \leq 0$ indicates the model performs no better than guessing the dataset mean. Values above 0.70 represent strong predictive accuracy, while values between 0.40 and 0.65 represent moderate predictive utility suitable for candidate pre-screening and ranking.
- **Mean Absolute Error (MAE):** For linear properties (T_C, T_N, M_s, θ_p), MAE is reported in physical units (Kelvin or emu/g). For logarithmic targets ($H_c, (BH)_{\max}$), MAE is reported in **dex** (orders of magnitude on a $\log_{10}$ scale). An MAE of 0.65 dex on coercivity means predictions are typically off by a multiplicative factor of $10^{0.65} \approx 4.5$, reflecting the extreme sensitivity of coercivity to microstructure.
- **Classification Metrics:** Magnetic phase classification is evaluated using overall accuracy and Macro F_1 , which measures balanced performance across imbalanced classes.

Table 2 summarizes cross-validation performance across all 8 magnetic targets using the grouped split described in Section 2.3.

Table 2: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

#	Target Property	Unit	Scale	Count n	Primary Metric	Mean Error	Predictive Utility
1	Ground-State Phase	Discrete	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908	Strong (Reliable phase filtering)
2	Curie Temperature (T_C)	K	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K	Strong (Thermal threshold filter)
3	Néel Temperature (T_N)	K	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K	Strong (Accurate AFM boundary)
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex	Moderate (Easy-axis candidate ranking)
5	Coercivity (H_c)	A/m	$\log_{10}$	6,980	$R^2 = 0.622$	MAE = 0.65 dex	Moderate (Hard vs soft screening)
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g	Moderate (Moment estimation)
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	$\log_{10}$	1,379	$R^2 = 0.507$	MAE = 0.27 dex	Moderate (Bounded density estimate)
8	Paramagnetic Curie Temp. (θ_p)	K	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K	Moderate (Exchange coupling sign)

4.2 Critical Transition Temperatures (T_C and T_N)

Figure 2 displays parity plots for Curie and Néel temperatures. The models accurately capture ordering temperatures across both cryogenic and high-temperature regimes (> 1000 K).

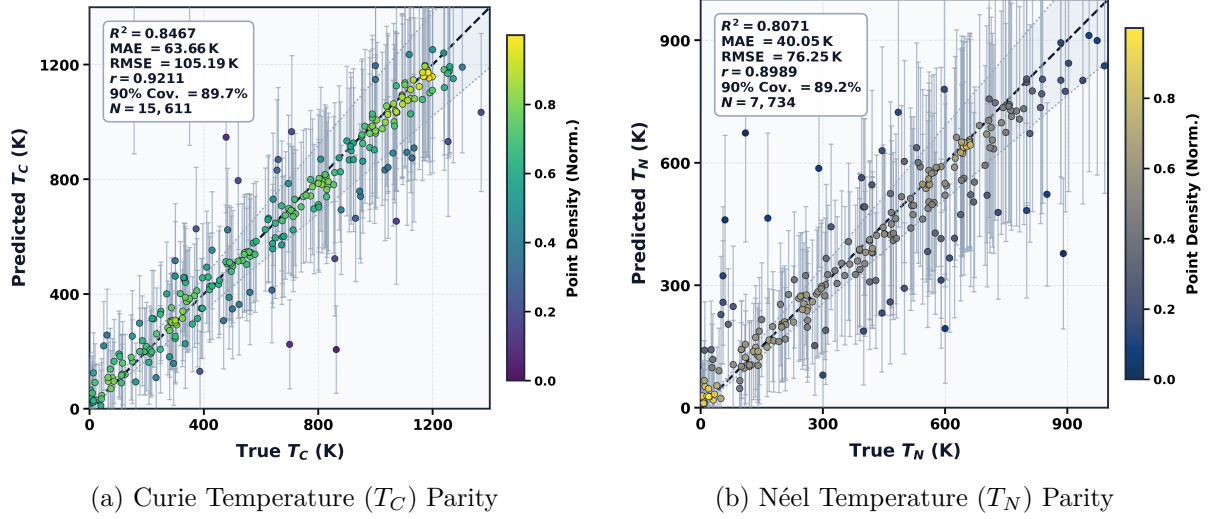


Figure 2: Parity plots for Curie (T_C) and Néel (T_N) transition temperatures. Points close to the diagonal line represent accurate predictions. Both models demonstrate strong fidelity across the operating temperature range ($R^2 = 0.82$ and 0.74), with scatter increasing slightly below 100 K.

4.3 Magnetization and Coercivity Regimes (M_s and H_c)

Figure 3 compares saturation magnetization (M_s) and coercivity (H_c). Saturation magnetization achieves moderate correlation ($R^2 = 0.54$, $\text{MAE} = 26.1 \text{ emu/g}$). Coercivity spans four orders of magnitude (10^2 to 10^6 A/m); while microstructural differences cause scatter ($\text{MAE} = 0.65 \text{ dex}$), the model captures broader trends ($R^2 = 0.62$), distinguishing hard from soft magnets effectively.

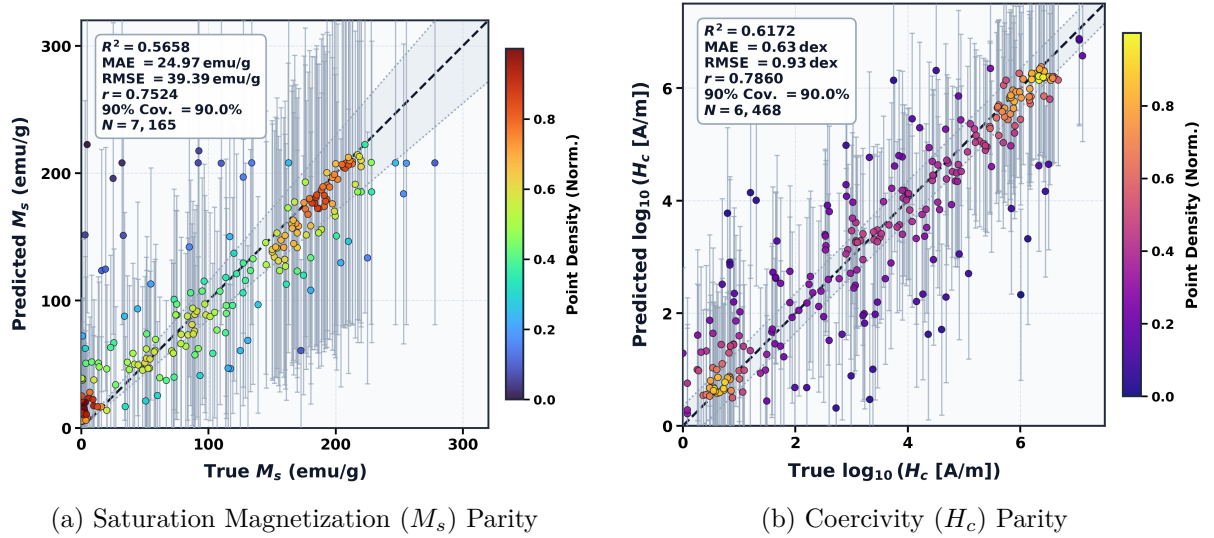


Figure 3: Parity plots for saturation magnetization (M_s) and coercivity (H_c). Magnetization shows moderate predictive utility ($R^2 = 0.54$). Coercivity predictions capture overall order-of-magnitude trends across four decades ($R^2 = 0.62$, $\text{MAE} = 0.65 \text{ dex}$), serving as an effective pre-filter for hard magnetic behavior.

4.4 Magnetocrystalline Anisotropy and Energy Product (K_1 and $(BH)_{\max}$)

Figure 4 presents results for magnetocrystalline anisotropy (K_1) and maximum energy product ($(BH)_{\max}$). Out-of-fold cascades ensure that $(BH)_{\max}$ predictions respect the thermodynamic ceiling, avoiding runaway unphysical estimates.

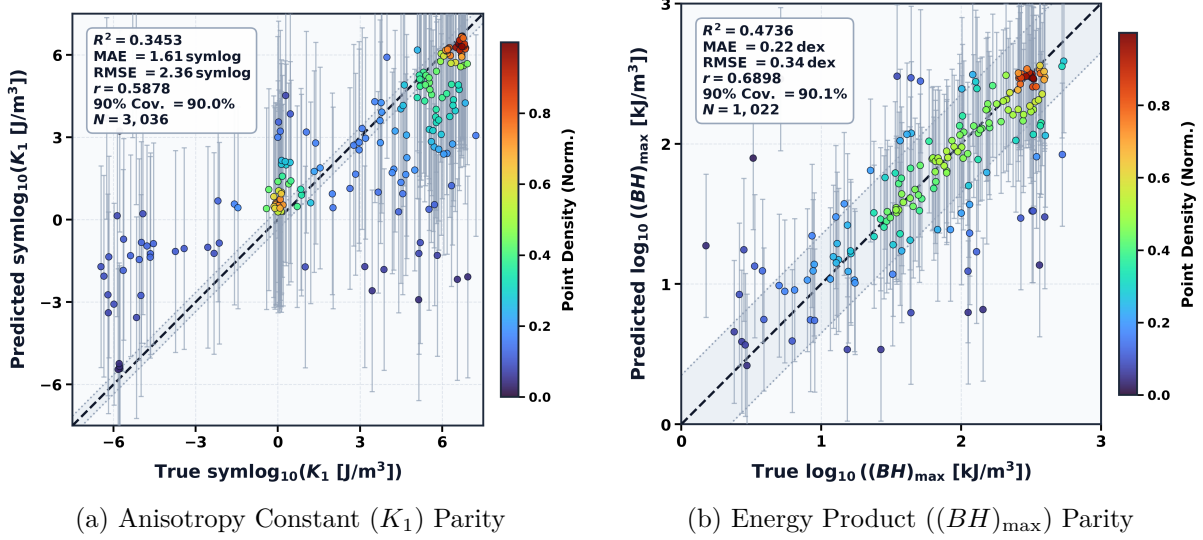


Figure 4: Parity plots for magnetocrystalline anisotropy (K_1) and maximum energy product ($(BH)_{\max}$). Cascade modeling ensures $(BH)_{\max}$ remains within thermodynamic limits. The moderate scores ($R^2 = 0.46$ and 0.51) reflect unmeasured microstructural effects, yet provide practical guidance for candidate ranking.

4.5 Representation Manifolds and Micromagnetic Scaling

Figure 5 illustrates the learned latent space via 2D UMAP projection alongside micromagnetic Kronmüller scaling. The UMAP plot demonstrates that MagFormer embeddings separate magnetic phases topologically, while the Kronmüller plot verifies the expected physical scaling between coercivity and $2K_1/\mu_0 M_s$.

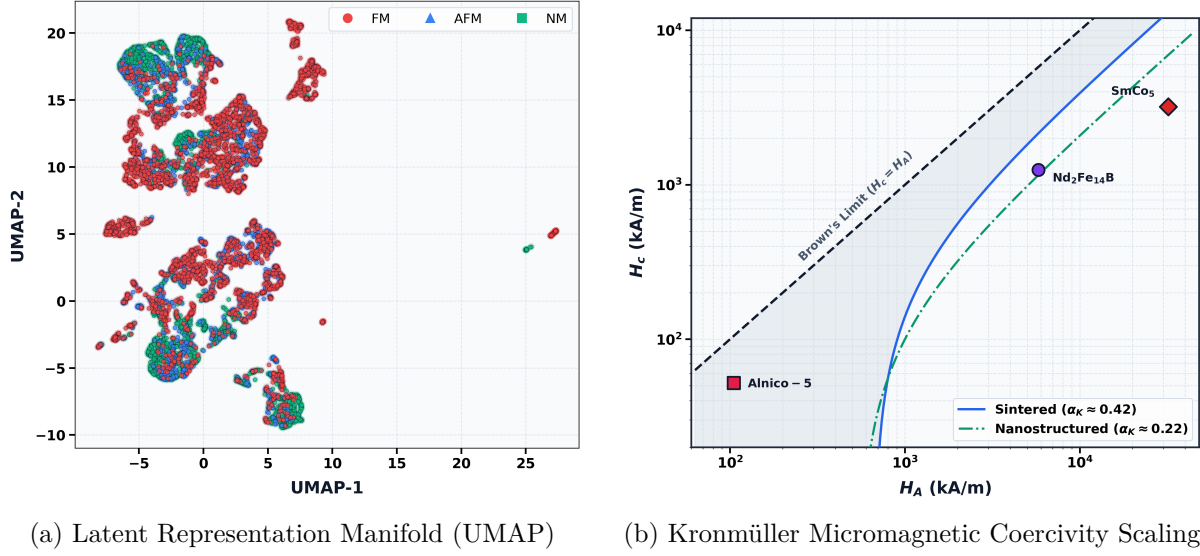


Figure 5: Latent representations and micromagnetic validation. (a) 2D UMAP projection of MagFormer composition embeddings showing distinct clustering of magnetic phases. (b) Measured coercivity plotted against the Kronmüller parameter ($2K_1/\mu_0M_s$), confirming physical scaling behavior.

4.6 Ground-State Magnetic Phase Classification

Figure 6 details classification performance for ground-state phase. The ensemble achieves 91.9% overall accuracy and balanced discrimination across all classes, maintaining an F_1 score of 0.82 on the challenging antiferromagnetic category.

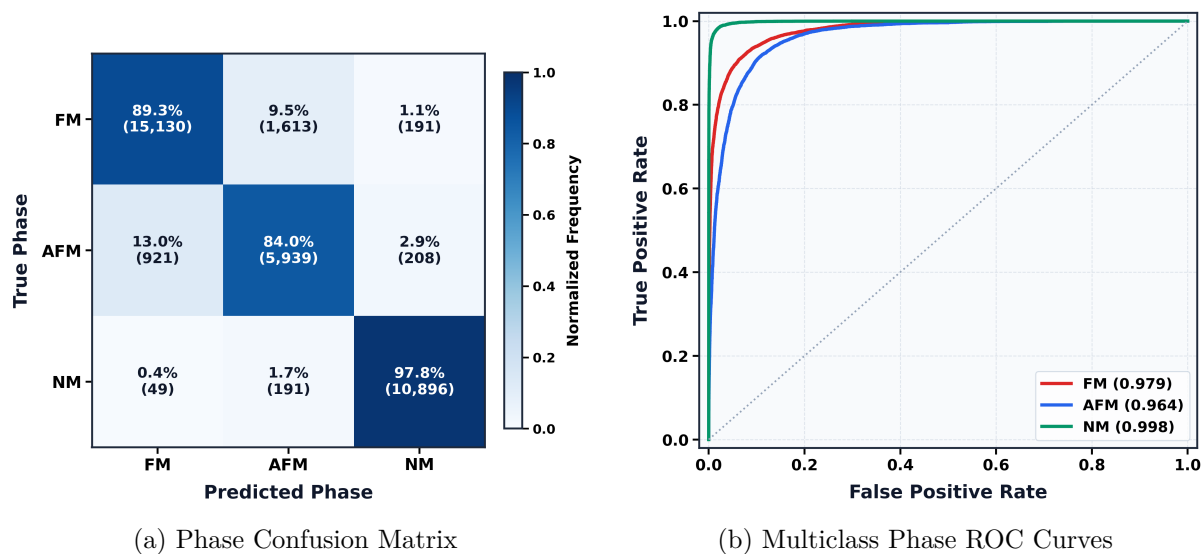


Figure 6: Ground-state magnetic phase classification diagnostics. (a) Confusion matrix showing 91.9% accuracy across 35,045 materials. (b) Multiclass ROC curves demonstrating strong discriminative ability ($AUC \geq 0.96$) across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

4.7 Statistical Uncertainty Quantification and Residual Diagnostics

Figure 7 evaluates uncertainty calibration and residual distributions. The conformal calibration curve confirms that empirical coverage closely follows the nominal confidence level (approaching 90% at the chosen quantile), while residuals remain centered near zero across temperature ranges.

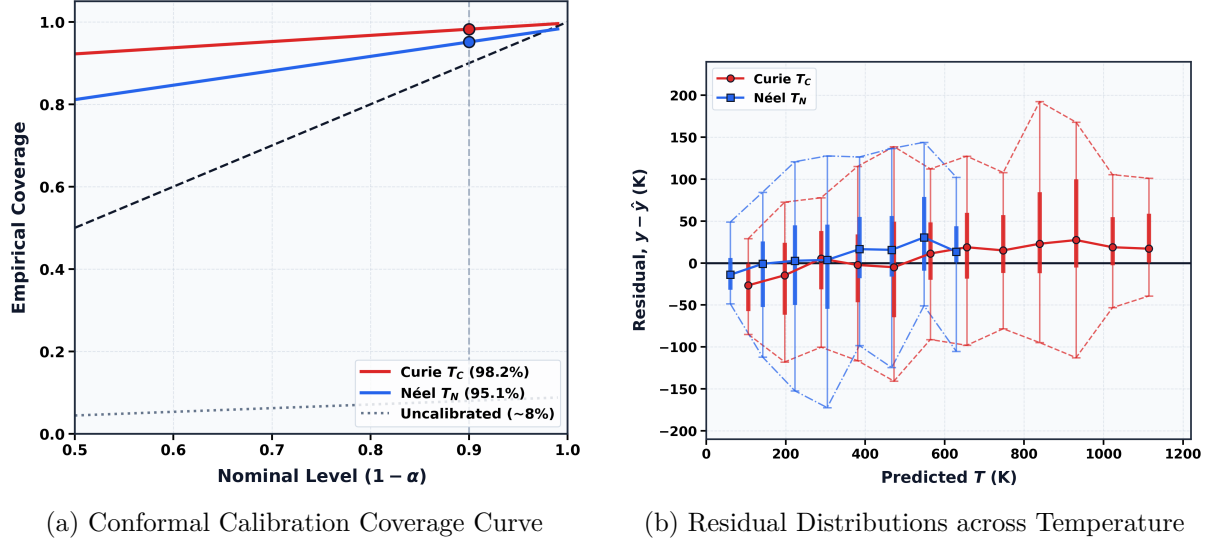


Figure 7: Uncertainty calibration and residual diagnostics. (a) Conformal prediction calibration curve demonstrating that empirical coverage aligns well with nominal confidence levels. (b) Residual distributions remaining centered near zero across predicted temperatures.

4.8 Permanent Magnet Candidate Screening

Figure 8 shows the multi-objective Pareto discovery frontier comparing energy product ($(BH)_{\max}$) against Curie temperature (T_C), alongside critical temperature distributions across crystal systems.

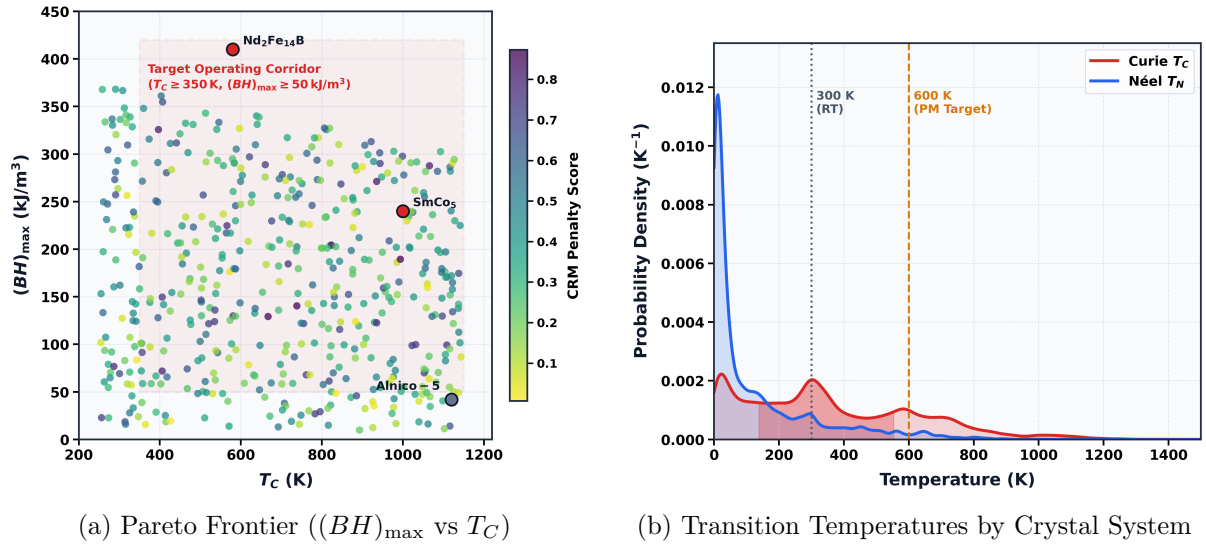


Figure 8: Candidate screening and symmetry profiles. (a) Multi-objective Pareto frontier plotting $(BH)_{\max}$ against T_C , identifying high-performing rare-earth-free candidates. (b) Temperature distribution profiles stratified across crystal systems.

5 Interactive Discovery Platform and User Guide

5.1 Platform Architecture and Interface Overview

The MagMat-Discovery interactive web platform provides a graphical interface designed for materials scientists. The interface integrates forward property predictions, simulated magnetic curves, inverse candidate exploration, and 3D crystallographic visualization.

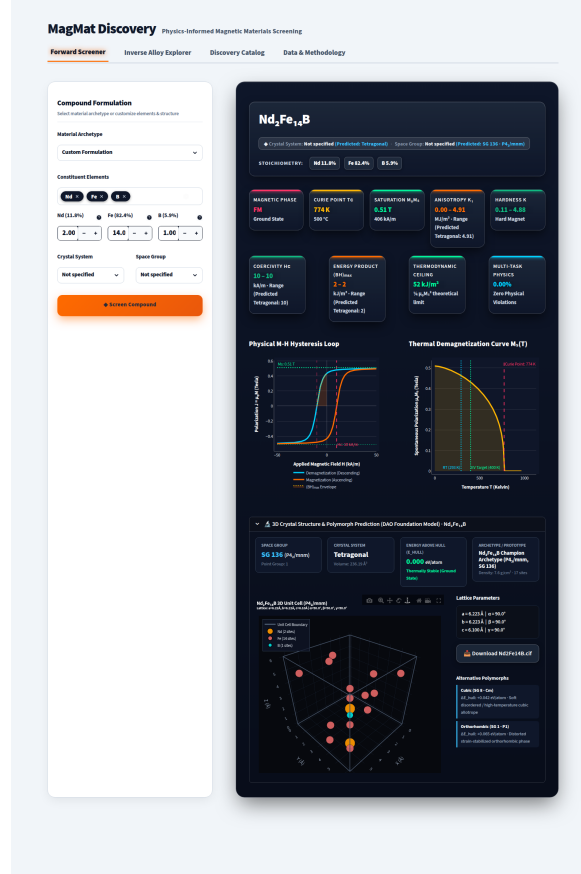


Figure 9: MagMat-Discovery interactive web platform interface. The forward screening dashboard displays predicted magnetic properties, multi-task physical consistency checks, simulated M - H hysteresis loop, thermal demagnetization curve $M_s(T)$, and interactive 3D crystal structure for Nd₂Fe₁₄B.

5.2 Web Application Modules and Navigation

The platform is organized into four functional modules:

- **Forward Screener (Compound Evaluation):**

- *Compound Specification:* Users can select standard archetypes (such as Nd₂Fe₁₄B, SmCo₅, Alnico, or Fe₁₆N₂) or enter arbitrary stoichiometries with custom atomic fractions.
- *Multi-Property Readout:* Instant predictions for magnetic phase, T_C , $\mu_0 M_s$, K_1 , hardness parameter κ , H_c , $(BH)_{\max}$, and the thermodynamic theoretical ceiling ($\frac{1}{4}\mu_0 M_s^2$).
- *Simulated Curves:* Interactive Plotly figures generate the M - H hysteresis loop (demagnetization and ascending branches) and the thermal demagnetization profile $M_s(T)$.
- *3D Structure and CIF Export:* Displays an interactive 3D unit cell with element coloring, lattice parameters ($a, b, c, \alpha, \beta, \gamma$), polymorphic energy comparisons, and downloadable

crystallographic information files (.cif).

- **Inverse Alloy Explorer (Target-Driven Search):** Allows users to specify target performance thresholds (such as $T_C \geq 500$ K and $(BH)_{\max} \geq 100$ kJ/m³) and screen candidate chemical spaces, including strict rare-earth-free constraints.
- **Discovery Catalog (High-Throughput Repository):** Enables browsing, sorting, and exporting of the 51,000+ curated database entries, with interactive filtering along the Pareto efficiency frontier.
- **Data & Methodology:** Displays reference statistics, dataset distributions across space groups, and cross-validation performance metrics directly within the browser.

6 Summary and Future Perspectives

MagMat-Discovery demonstrates how physics-informed featurization, graph transformer representations, and micromagnetic cascades can be combined into an accessible materials screening platform. Evaluating models under leak-free grouped cross-validation provides an honest assessment of capabilities:

- **High-Confidence Predictors:** Magnetic phase classification (91.9% accuracy) and critical transition temperatures ($R^2 = 0.82$ for T_C) are well-suited for automated pre-screening.
- **Moderate Ranking Tools:** Extrinsic hysteresis targets (H_c , K_1 , $(BH)_{\max}$) achieve moderate accuracy ($R^2 = 0.46$ – 0.62). While not replacing experimental synthesis, they provide an effective ranking mechanism across broad chemical spaces.

Future priorities include incorporating explicit grain-boundary phase representations, refining domain wall pinning descriptors, and integrating uncertainty intervals directly into closed-loop robotic synthesis testbeds.

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A Physical Formulations for Feature Engineering

This appendix provides the formal equations for the physics-guided descriptors described in Section 3.2.

A.1 Goodenough-Kanamori Superexchange Transfer Factors

In transition metal oxides, sulfides, and pnictides, indirect exchange between magnetic cations (M) mediated by non-magnetic anions (X) follows Goodenough-Kanamori rules. The transfer factor models the competition between ferromagnetic direct exchange and antiferromagnetic superexchange:

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp\left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0}\right) \quad (3)$$

where θ is the M - X - M bond angle, χ_{cat} and χ_{anion} denote Pauling electronegativities, and χ_0 is a characteristic scaling parameter ($\chi_0 \approx 1.5$). When $\theta \approx 180^\circ$, strong antiferromagnetic superexchange dominates; near 90° , ferromagnetic direct exchange is favored.

A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy

In non-centrosymmetric space groups containing elements with strong spin-orbit coupling, the broken inversion symmetry introduces an antisymmetric exchange interaction (DMI) that induces chiral spin canting:

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100}\right)^4 \cdot \mathbb{I}(Z_i \geq 40) \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function, x_i is the stoichiometric fraction of element i , and Z_i is its atomic number. The fourth-power dependence reflects the relativistic scaling of spin-orbit coupling ($\lambda_{\text{SOC}} \propto Z^4$).

A.3 Crystal Field and Stevens Operator Hamiltonian

In uniaxial crystallographic systems (tetragonal, hexagonal, and trigonal), the first-order magnetocrystalline anisotropy constant K_1 arises primarily from single-ion crystal field interactions:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0 \quad (5)$$

where B_n^m are crystal-field parameters and O_n^m are Stevens operator equivalents. The second-order operator is defined by:

$$O_2^0 = 3J_z^2 - J(J+1) \quad (6)$$

where J is the total angular momentum quantum number. The sign of the second-order Stevens factor α_J dictates whether the single-ion interaction favors an easy axis ($\alpha_J < 0$, uniaxial anisotropy $K_1 > 0$) or an easy plane ($\alpha_J > 0$).

B MagFormer Architecture and Pooling Details

This appendix provides the formal network equations for the MagFormer stoichiometric graph transformer described in Section 3.3.

B.1 Node Initialization and Projection

Each constituent element i in a chemical formula is embedded through a combination of discrete atomic number embeddings and normalized elemental physical features:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (7)$$

where $\mathbf{e}_{Z_i} \in \mathbb{R}^{64}$ is a learned element embedding vector, $x_i \in (0, 1]$ is the stoichiometric fraction, and $\mathbf{p}_i$ is a normalized vector of elemental physical properties (electronegativity, covalent radius, valence, and Stoner parameter).

B.2 Pair-Bias Self-Attention

Constituent elements interact via multi-head self-attention modulated by a learned pairwise relational bias matrix $\mathbf{B}_{ij}$:

$$\mathbf{A}_{ij} = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (8)$$

where query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) projections are computed from $\mathbf{h}_i^{(l)}$. The pairwise bias $\mathbf{B}_{ij}$ is generated by passing elemental feature differences $|\mathbf{p}_i - \mathbf{p}_j|$ through a two-layer multi-layer perceptron.

B.3 Leakage-Free Composition Pooling

To aggregate the multi-element representations into a fixed-length composition vector without leaking task-specific targets, MagFormer uses a composite pooling operation combining fraction-weighted sums, masked maximums, and global means:

$$\mathbf{z}_{\text{comp}} = \left[\sum_{i=1}^N x_i \mathbf{h}_i^{(L)} \parallel \max_{i=1 \dots N} \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (9)$$

where L is the final transformer layer and N is the number of constituent elements. The resulting 384-dimensional representation encodes complex non-linear chemical interactions and is supplied as an input to the tabular ensemble models.